

QI ZHAO

130 Bd de l'Europe, 76100 Rouen, France | 26 years old | +33 06 27 03 96 33 | Qi.zhao@groupe-esigelec.org

Medical systems engineer trained in the ISYMED (Medical Systems Engineering) track at ESIGELEC, with hands-on research experience applying deep learning to clinical imaging. Recent work at the Computer Vision Laboratory of the University of Würzburg focused on diffusion-model based SPECT image enhancement and multi-sensor synchronization for motion analysis.

WORK EXPERIENCE

Medical Image Enhancement by Diffusion Model05/2025 – 08/2025 · Würzburg
Computer Vision Laboratory, University of Würzburg

- Built a complete 3D medical-image enhancement framework integrating diffusion model DDIM with an existing clinical SPECT reconstruction pipeline, in collaboration with Universitätsklinikum Würzburg, using their provided papers, databases, and R2U-Net model.
- Modified the existing R2U-Net into a dual-channel structure to optimize SPECT image detail and blur recognition, while preserving the original architecture and evaluation flow; integrated diffusion models with the existing PyTorch Lightning training infrastructure for improved adaptability.

Multi-Event Camera Synchronization Framework03/2025 – 04/2025 · Würzburg
Computer Vision Laboratory, University of Würzburg

- Designed and implemented a multi-camera data acquisition framework enabling simultaneous control of heterogeneous imaging sensors, including DAVIS346 event cameras and DVSense devices, with synchronized triggering, real-time previews, and structured data storage for downstream motion and scene-change analysis.

ACADEMIC PROJECTS

CNN for Pneumonia Classification11/2024 – 01/2025 · Rouen

- Trained a CNN on a chest X-ray dataset with data augmentation to improve robustness, diagnostic accuracy, and speed; documented results in an English paper written in Overleaf.

First Prize, Hackathon — Implantable Inertial Unit10/2024 – 11/2024 · Rouen

- Developed an implantable motion-tracking system for patients with significant loss of motor function, tracking neighboring muscle position/movement cycles; awarded first prize.

Facial Detection Model02/2024 – 04/2024 · Rouen

- Built a 2D/3D face detection pipeline based on MobileNetV2, using the FDDB dataset to recognize and frame known/unknown faces in photos and video streams.

Robot Competition — Locomotive & Follower01/2024 – 02/2024 · Rouen

- Designed two coordinated robots — a line-following locomotive (US/IR sensors) and an IR-guided follower — programmed on an MSP430 microcontroller.

EDUCATION

ISYMED Dominant Engineering Diploma — Medical Systems Engineering
2022 · Rouen

ESIGELEC

- Robotique chirurgicale et téléopération: researched and presented papers on medical technologies and AI; final assignment was a report on using CNN for Pneumonia Classification.
- ES Artificial Intelligence for Healthcare, Level 1 & 2: deep learning techniques, data analysis methods, and AI model evaluation for medical diagnostics and treatment.
- Notions générales en Machine Learning: core ML algorithms, model training, feature selection, and data preprocessing.
- Management et plateaux techniques hospitaliers: visited local hospitals in Rouen to study sterilization/maintenance of surgical equipment and drug distribution management.
- Dispositif médical — impact réglementaire: regulatory considerations for embedded medical devices; studied modification, testing, and validation of an automated injection system.

COMPUTER SKILLS

Python
CUDA
OpenCV
Jupyter
PyTorch
Arduino
MATLAB
Linux

LANGUAGES

Chinese
Native
English
TOEIC 860 (B2)
French
DELFB2

ASSETS

Perseverance
Passionate
Teamwork